

Interacting Multipoles of Rank 3 and 3:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta} = R_z^{-13} \cdot -\frac{1}{945}.$$

[illegible]

Simplify deltas

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta} = R_z^{-13} \cdot -\frac{1}{945}.$$

[illegible]

Simplify R

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta} = R_z^{-13} \cdot -\frac{1}{945}.$$

[illegible]

Simplify Multipoles

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta}=R_z^{-13}\cdot-\frac{1}{945}\cdot$$

Exploit Traceless Conditions

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta}=R_z^{-13}\cdot-\frac{1}{945}\cdot$$

$$6$$

$$\begin{aligned}
&+105\cdot R_z^6\cdot\Omega_{z\alpha\gamma}\cdot\Omega_{z\alpha\gamma}\\
&+105\cdot R_z^6\cdot\Omega_{z\alpha\beta}\cdot\Omega_{z\alpha\beta}\\
&+105\cdot R_z^6\cdot\Omega_{z\alpha\beta}\cdot\Omega_{z\alpha\beta}\\
&+105\cdot R_z^6\cdot\Omega_{z\alpha\beta}\cdot\Omega_{z\alpha\beta}\\
&+105\cdot R_z^6\cdot\Omega_{z\alpha\beta}\cdot\Omega_{z\alpha\beta}\\
&+105\cdot R_z^6\cdot\Omega_{z\alpha\beta}\cdot\Omega_{z\alpha\beta}\\
&+105\cdot R_z^6\cdot\Omega_{z\alpha\beta}\cdot\Omega_{z\alpha\beta}\\
&+105\cdot R_z^6\cdot\Omega_{zz\gamma}\cdot\Omega_{\alpha\alpha\gamma}\\
&+105\cdot R_z^6\cdot\Omega_{zz\beta}\cdot\Omega_{\alpha\alpha\beta}\\
&+105\cdot R_z^6\cdot\Omega_{zz\alpha}\cdot\Omega_{\alpha\beta\beta}\\
&+105\cdot R_z^6\cdot\Omega_{zz\gamma}\cdot\Omega_{\alpha\alpha\gamma}\\
&+105\cdot R_z^6\cdot\Omega_{zz\beta}\cdot\Omega_{\alpha\alpha\beta}\\
&+105\cdot R_z^6\cdot\Omega_{zz\alpha}\cdot\Omega_{\alpha\beta\beta}\\
&+105\cdot R_z^6\cdot\Omega_{zz\gamma}\cdot\Omega_{\alpha\alpha\gamma}\\
&+105\cdot R_z^6\cdot\Omega_{zz\beta}\cdot\Omega_{\alpha\alpha\beta}\\
&+105\cdot R_z^6\cdot\Omega_{zz\alpha}\cdot\Omega_{\alpha\beta\beta}\\
&-15\cdot R_z^6\cdot\Omega_{\alpha\alpha\gamma}\cdot\Omega_{\gamma\epsilon\epsilon}\\
&-15\cdot R_z^6\cdot\Omega_{\alpha\alpha\gamma}\cdot\Omega_{\gamma\delta\delta}\\
&-15\cdot R_z^6\cdot\Omega_{\alpha\alpha\gamma}\cdot\Omega_{\gamma\delta\delta}\\
&-15\cdot R_z^6\cdot\Omega_{\alpha\alpha\beta}\cdot\Omega_{\beta\epsilon\epsilon}\\
&-15\cdot R_z^6\cdot\Omega_{\alpha\alpha\beta}\cdot\Omega_{\beta\delta\delta}\\
&-15\cdot R_z^6\cdot\Omega_{\alpha\alpha\beta}\cdot\Omega_{\beta\delta\delta}\\
&-15\cdot R_z^6\cdot\Omega_{\alpha\beta\beta}\cdot\Omega_{\alpha\epsilon\epsilon}\\
&-15\cdot R_z^6\cdot\Omega_{\alpha\beta\gamma}\cdot\Omega_{\alpha\beta\gamma}\\
&-15\cdot R_z^6\cdot\Omega_{\alpha\beta\gamma}\cdot\Omega_{\alpha\beta\gamma}
\end{aligned}$$

$(5a)$

Reduce Indices

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta} = R_z^{-13} \cdot -\frac{1}{945}.$$

7

[illegible]

Add up

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta} = R_z^{-13} \cdot -\frac{1}{945} \cdot \left[\begin{array}{l} -2835 \cdot R_z^6 \cdot \Omega_{zz\gamma} \cdot \Omega_{zz\gamma} \\ -2835 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{zz\beta} \\ -2835 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{zz\alpha} \\ +315 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\beta\beta} \\ +315 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\gamma\gamma} \\ +630 \cdot R_z^6 \cdot \Omega_{z\alpha\delta} \cdot \Omega_{z\alpha\delta} \\ +315 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{\beta\gamma\gamma} \\ +630 \cdot R_z^6 \cdot \Omega_{z\beta\delta} \cdot \Omega_{z\beta\delta} \\ +630 \cdot R_z^6 \cdot \Omega_{z\gamma\delta} \cdot \Omega_{z\gamma\delta} \\ +315 \cdot R_z^6 \cdot \Omega_{zz\delta} \cdot \Omega_{\beta\beta\delta} \\ +315 \cdot R_z^6 \cdot \Omega_{zz\delta} \cdot \Omega_{\gamma\gamma\delta} \\ +315 \cdot R_z^6 \cdot \Omega_{zz\gamma} \cdot \Omega_{\gamma\delta\delta} \\ -45 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\beta\beta\gamma} \\ -45 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\beta\gamma\gamma} \\ -45 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta} \cdot \Omega_{\alpha\gamma\gamma} \\ -90 \cdot R_z^6 \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\beta\gamma} \end{array} \right] \quad (7)$$

Combining everything

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta} = \left[\begin{array}{l} +3 \cdot R_z^{-7} \cdot \Omega_{zz\gamma} \cdot \Omega_{zz\gamma} \\ +3 \cdot R_z^{-7} \cdot \Omega_{zz\beta} \cdot \Omega_{zz\beta} \\ +3 \cdot R_z^{-7} \cdot \Omega_{zz\alpha} \cdot \Omega_{zz\alpha} \\ -1/3 \cdot R_z^{-7} \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\beta\beta} \\ -1/3 \cdot R_z^{-7} \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\gamma\gamma} \\ -2/3 \cdot R_z^{-7} \cdot \Omega_{z\alpha\delta} \cdot \Omega_{z\alpha\delta} \\ -1/3 \cdot R_z^{-7} \cdot \Omega_{zz\beta} \cdot \Omega_{\beta\gamma\gamma} \\ -2/3 \cdot R_z^{-7} \cdot \Omega_{z\beta\delta} \cdot \Omega_{z\beta\delta} \\ -2/3 \cdot R_z^{-7} \cdot \Omega_{z\gamma\delta} \cdot \Omega_{z\gamma\delta} \\ -1/3 \cdot R_z^{-7} \cdot \Omega_{zz\delta} \cdot \Omega_{\beta\beta\delta} \\ -1/3 \cdot R_z^{-7} \cdot \Omega_{zz\delta} \cdot \Omega_{\gamma\gamma\delta} \\ -1/3 \cdot R_z^{-7} \cdot \Omega_{zz\gamma} \cdot \Omega_{\gamma\delta\delta} \\ +1/21 \cdot R_z^{-7} \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\beta\beta\gamma} \\ +1/21 \cdot R_z^{-7} \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\beta\gamma\gamma} \\ +1/21 \cdot R_z^{-7} \cdot \Omega_{\alpha\beta\beta} \cdot \Omega_{\alpha\gamma\gamma} \\ +2/21 \cdot R_z^{-7} \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\beta\gamma} \end{array} \right] \quad (8)$$